

Whether Token Probabilities Catch a Wrong Tool Call Depends on the Model and the Kind of Error

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Abstract

Teams that deploy tool-calling language models want to stop a wrong call before it executes, using the token probabilities every serving stack returns or a probe on the model’s hidden states. Earlier comparisons pooled every kind of error and held out items rather than tools, so they cannot say which wrong calls the cheaper signal misses. We compare the two on held-out tools for each kind of error, on six open-weight checkpoints of at most three billion parameters from four developers, beside a length floor and two judges that read only the output. On wrong argument values, the probe ranks errors above valid calls better than the mean token log-probability on six of seven Llama-3.2 and Gemma-3 runs, by +0.134 to +0.357 in area under the curve, and worse on all six Qwen3, Qwen3.5 and MiniCPM5 runs, including the runs forced to write JSON. Much of the probe’s lead on value errors is readable from the output: a judge trained on the output text recovers at least half of it on three of the six runs where the probe leads, and none of it on Llama-3.2-3B. The sides are assigned by family and hold three checkpoints each, so the split is a description, and before building an internal judge an operator should score the model’s confidence on its value errors and train a judge on the output text first.

1. Introduction

A language model agent acts by calling tools. When it calls the wrong tool, or the right one with wrong arguments, the call is usually well formed, so the runtime executes it before anyone reads the transcript. A guardrail therefore has to judge the call between its generation and its execution. Two signals are available there. The output distribution gives the

probability of each generated token, which every serving stack returns. The hidden states give a probe something to read, which only a stack that runs the model’s own code can expose.

Probes on the residual stream separate wrong calls from right ones with high accuracy (Healy et al., 2026; Yeats et al., 2026), and Ke et al. (2026) separate three failures, invalid arguments, unneeded calls and missing calls, and detect each from a few neurons. Noël et al. (2026) compared token probabilities with probes on tool calls under splits by prompt. What these studies do not report is which wrong calls the model’s own token probabilities miss on tools absent from training, and whether a judge that reads only the output text would catch them. That is the question this paper answers.

We answer it with one quantity, the internal advantage: the area under the curve (AUC) of a probe on the residual stream minus that of the mean token log-probability of the call, on tools never seen in training, beside a floor that reads only lengths. We compute it on every scored failure and on each kind of error against valid calls, and we set the probe against two trained comparators that need no access to the deployed model’s states: an output-only judge fitted on the same labels and a second small language model that reads the request and the call.

The main result is the value-error split (Figure 1b). On wrong argument values the probe leads confidence on six of the seven Llama-3.2 and Gemma-3 runs, by +0.134 to +0.357 (point estimates, intervals excluding zero on five). Confidence leads the probe on all six Qwen3, Qwen3.5 and MiniCPM5 runs, native or forced into JSON, with intervals excluding zero on three. The two sides are assigned by family and hold three checkpoints each, so the split describes these checkpoints and is not a test. Much of the probe’s lead on value errors is readable from the output: the output-only judge recovers at least half of it on three of the six probe-side runs where the probe leads, and none of it on either Llama-3.2-3B run.

In practice this gives an order of operations (Section 8): score the model’s confidence on its value errors on held-out tools first, where it misses them train a judge on the output text and compare it with a probe before choosing, and for

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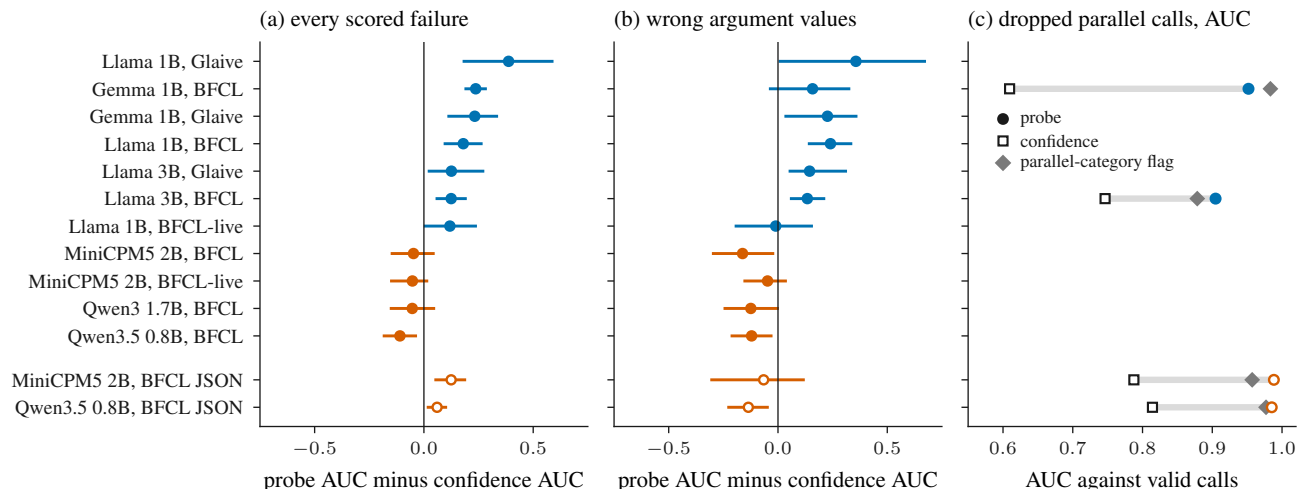


Figure 1. (a) Token-role probe AUC minus mean log-probability AUC on held-out tools, every scored failure against valid calls, dot at the mean over five split seeds, bar the 95% interval from resampling tools. (b) The same on wrong argument values against valid calls, for runs with at least twenty of them. (c) AUC of the probe (dot), the mean log-probability (open square) and a one-bit flag for a request in a parallel category (grey diamond) on dropped parallel calls against valid calls, for runs with at least twenty dropped calls. Blue Llama-3.2 and Gemma-3, orange Qwen3, Qwen3.5 and MiniCPM5, open markers the forced-JSON runs.

parallel requests count the calls, since a one-bit category flag already separates dropped calls on these runs.

Contributions.

1. The value-error split, measured on 11 powered runs from six checkpoints of four developers on held-out tools, with a length floor and a paired interval on every difference (Section 4). This paper adds the decomposition by kind of error against the model’s own confidence on held-out tools. Ke et al. (2026) split failures into validity, over-calling and missing calls for internal detectors, without the model’s confidence beside them.
2. The comparators that bound the claim: an output-only judge and a reader model under the probe’s protocol, on every scored failure and on value errors, which show on which runs the probe’s lead is readable from the text (Section 5).
3. The controls and the disclosures that condition the split: item difficulty, label budget, the confidence summary, duplicated Glaive requests, the prompt route, extraction provenance, and a dropped-call contrast that a one-bit category flag matches (Sections 4 and 6).

2. Related work

Probes on hidden states. Probes on the residual stream recover truthfulness (Azaria & Mitchell, 2023; Marks & Tegmark, 2024), hallucination risk (Ji et al., 2024) and sampling-based uncertainty in one pass (Kossen et al., 2024). Orgad et al. (2025) locate the signal at the exact-answer tokens, which is the observation the token-role probe of Healy

et al. (2026) applies to a call’s name, arguments and closing delimiter. Yeats et al. (2026) train linear probes on eighteen models on BFCL, find that model size predicts probe accuracy first and probing depth second, and find that tool-specific fine-tuning lowers it. Ke et al. (2026) split tool-use failures into invalid arguments, over-calling and missing calls and detect each from a few neurons on six models. These results establish that internal detectors of wrong calls work. Healy et al. (2026) and Yeats et al. (2026) hold out items, so a tool seen in training can return at test, and none of the three reports the model’s own token probabilities on the same items.

Token probabilities against probes on tool calls. Noël et al. (2026) compared the mean token log-probability with the token-role probe and with head-averaged spectral probes on Glaive under splits by prompt, and found the log-probability near chance. Under tool-held-out splits and repaired labels that finding does not hold here: confidence ranges from 0.576 to 0.823 AUC on the powered runs (Section 4).

Readouts of attention. LapEigvals trains a probe on the top eigenvalues of the attention Laplacian (Binkowski et al., 2025), Lookback Lens on the share of attention paid to the context (Chuang et al., 2024), LLM-Check on eigenvalues of attention kernels (Sriramanan et al., 2024) and SinkProbe on attention sinks (Binkowski et al., 2026). Noël (2025) introduced graph-signal diagnostics on a head-averaged attention graph, and Dahlem et al. (2026) prove that transpose-invariant spectral diagnostics are blind to the direction of attention flow. We reimplement LapEigvals, Lookback Lens

and SinkProbe after their authors’ papers and code. No numerical parity check against the original implementations is reported, so their numbers here are those of our reimplementations.

Output confidence. Token probabilities are informative about a model’s own correctness on question answering (Kadavath et al., 2022), and sampling-based estimates sharpen them (Farquhar et al., 2024; Manakul et al., 2023) at a cost in forward passes that an agent about to act does not have. Tool-calling benchmarks measure whether a call is right (Patil et al., 2025; 2024; Qin et al., 2024) and not whether the model’s confidence tracks it.

3. Setup

Task, labels and kinds of error. A model receives a user request and a set of tool schemas and generates greedily. The call is parsed in whichever format the model writes and compared with the benchmark’s ground truth after both are normalised to a name and an argument dictionary. Each call receives one failure mode: valid, wrong name, wrong argument values, missing arguments, missing parallel calls, an unneeded call, no call, an unparseable call, or a call cut off by the budget. Every number is computed on the scored population, the items where a call is required and the generation can be judged, and “every scored failure” means every failure mode in that population. We also score two kinds of error against the valid calls alone: wrong argument values, where the right tool gets a wrong value, and dropped parallel calls, where a request needs several calls and the model writes fewer.

Prompt route. Tool schemas reach the model through its own chat template where the template carries them. Gemma-3’s template drops the tool argument, so every Gemma-3 run, and both forced-JSON runs, use a fallback system prompt that lists the schemas and asks the model to reply with only one JSON object of the form name and arguments and nothing else (Section B). Rendering one prompt per model on CPU with the paper’s render function confirms that Gemma-3 is the only one of the six whose template drops the schemas. A prompt that asks for one object invites a model to drop the other calls of a parallel request, so dropped calls on these runs are partly a property of the prompt, and Section 4 treats them that way.

Conditioning facts. The labelling and extraction code was audited and repaired before the measurements here (Section C). The largest repair concerns correct list-valued arguments, which had been scored wrong on the JSON formats: on Qwen3-1.7B it turned 158 labelled failures into 52. Relabelling the stored generations of 16 runs with the released labeller reproduces every stored label. Two

extractor generations coexist (Table 5): five of the seven probe-side runs and one of the four confidence-side runs keep the first extraction, which records no commit, and the re-extracted runs do not all come from a clean tree. One run extracted under both reads +0.224 and +0.180, a drift of 0.044. Some failures are schema echoes, calls whose arguments copy the tool’s JSON schema: a pattern for a copied `properties` field or object type finds them in 73% of Gemma-3 failures on BFCL and on no confidence-side run. Most are labelled missing arguments and stay out of the value-error comparison.

Models and benchmarks. The checkpoints are Llama-3.2-1B and 3B, Gemma-3-1B, Qwen3-1.7B, Qwen3.5-0.8B and MiniCPM5-2B, from four developers, released between September 2024 and September 2026 (Section M). The chat templates of the Qwen3, Qwen3.5 and MiniCPM5 checkpoints carry a reasoning mode, which is disabled at generation throughout. The benchmarks are the Glaive function-calling corpus (GlaiveAI, 2024), the single-turn categories of BFCL-v4 (Patil et al., 2025) and its live categories, contributed by users. Glaive repeats requests: on Llama-3.2-1B its 580 scored items hold 286 unique requests. A run is one checkpoint on one benchmark. A run with fewer than thirty items of either class is underpowered and excluded from every aggregate, which leaves the confidence-side checkpoints with powered runs on BFCL and BFCL-live only. A kind of error is scored on a run only with at least twenty failures of that kind.

Detectors. Confidence is the mean token log-probability over the generated call, with its direction fixed on training items, chosen before any result was seen. It averages over every token of the call, while the probe reads the name, the argument values and the delimiter separately. The probe is the token-role probe of Healy et al. (2026), a regularised logistic regression on the residual state at the function name, the mean over argument-value tokens and the closing delimiter, at eight depths. The output-only judge is one logistic regression on what an operator sees from outside the model: lengths, benchmark category, counts read off the call, every stored confidence summary, and character n-grams of the call and the request. The reader is Qwen3.5-0.8B, which reads the request and the call as plain text, without schemas, and is probed with the same classifier. It is the same checkpoint as one confidence-side model. The floor is a classifier on prompt length, generation length and truncation.

Protocol and the two sides. Items are grouped by ground-truth tool and split five ways at the group level, so every score is on tools absent from training. Every trained detector is fitted on the training folds restricted to the scored population, with a validation carve-out of training tools. One pooled AUC is computed per split seed, and five seeds are

averaged. Every difference between detectors is a paired bootstrap that resamples whole tools, with its 95% percentile interval. The internal advantage of a run is the AUC of the probe minus that of confidence. Each run is assigned to a side by its family, fixed in the analysis code when the split was first described: Llama-3.2 and Gemma-3 on the probe side, Qwen3, Qwen3.5 and MiniCPM5 on the confidence side. The assignment came from the data it describes, and a run can disagree with its side.

4. The internal advantage by kind of error

On every scored failure, the seven powered Llama-3.2 and Gemma-3 runs have an internal advantage of +0.119 to +0.387, with an interval excluding zero on six of them (Table 1, Figure 1a). The four powered Qwen3, Qwen3.5 and MiniCPM5 runs read -0.110 to -0.048, with an interval below zero on one. Runs that share a checkpoint are not independent. With three against three checkpoints the smallest attainable two-sided rank-test value is $p = 0.100$, so the separation is a description of these six checkpoints.

Wrong argument values. Every scored failure mixes kinds of error, so Figure 1b scores wrong argument values against valid calls alone. The probe leads confidence on six of the seven probe-side runs, by +0.134 to +0.357 in point estimate, with an interval excluding zero on five. The largest lead, Llama-3.2-1B on Glaive, has the widest interval, [+0.00, +0.68]. Llama-3.2-1B on the live categories reads -0.010, on the probe side by family and on neither side by measurement. Confidence leads the probe on all six confidence-side runs, native and forced into JSON, where the advantage is -0.162 to -0.048, with an interval excluding zero on three. On these models the mean log-probability ranks most wrong values below right ones. Glaive repeats requests, and keeping the first occurrence of each unique request leaves the value-error advantage at +0.360 on Llama-3.2-1B and +0.235 on Gemma-3.

Dropped parallel calls. A one-bit flag for a request in a parallel category separates dropped calls from valid calls at 0.878 to 0.983 AUC on the four runs with at least twenty of them (Figure 1c), because only parallel requests can drop a call. On Gemma-3 the flag reaches 0.983 against the probe's 0.952. Three of these runs use the fallback prompt that asks for one JSON object. Within the parallel categories, where the flag is constant, the probe leads confidence on MiniCPM5-2B and Qwen3.5-0.8B under forced JSON, +0.330 [+0.20, +0.45] and +0.198 [+0.09, +0.32]. On Llama-3.2-3B, the only one of the four prompted through its native template, the lead vanishes at +0.016 [-0.10, +0.13], and Gemma-3 makes too few valid parallel calls to be tested. No claim about dropped calls is made beyond these three numbers.

Output format and R3. Qwen3 writes JSON like Llama-3.2 and Gemma-3 and sits on the confidence side, so the native format does not decide the side. Registered prediction R3 said that, forced to write JSON, the confidence-side families keep an advantage at or below zero on every scored failure, and that it fails above +0.10. It failed on MiniCPM5-2B, at +0.125 [+0.05, +0.19], on a partial run of 616 of 850 items extracted from a tree with uncommitted changes. On Qwen3.5-0.8B it is undecided by the rule, at +0.060 [+0.01, +0.11]. Dropped parallel calls make up 56 of 83 and 143 of 226 failures under the fallback prompt, against 16 and 19 under the native templates. On wrong argument values both models stay on the confidence side.

Size and release date. Size does not separate the sides (one and three billion parameters against 0.8 to two), and within Llama-3.2 the probe does not improve from 1B to 3B (0.831 at both sizes on BFCL). Release date, a reasoning mode and native tool-calling post-training all separate the six checkpoints (Section 8).

Calls with a history. One multi-turn run, MiniCPM5-2B, reads 0.900 for the probe and 0.859 for confidence, the opposite order to its single-turn runs, on four linked groups. It is anecdotal and enters no claim (Section L).

5. What the probe adds over the output text

The internal advantage sets a trained probe against an untrained summary of the output, so part of the probe's lead could come from training on labels rather than from the hidden states. The comparators of this section are trained on the same labels, folds and training population as the probe and read only what is visible outside the deployed model.

On value errors. Table 2 scores both comparators on wrong argument values against valid calls, the contrast that defines the split. The output-only judge recovers at least half of the probe's lead over confidence on three of the six probe-side runs where the probe leads (76% and 64% on Llama-3.2-1B, 94% on Gemma-3 on Glaive) and none of it on Llama-3.2-3B, where the probe beats it by +0.225 on Glaive and +0.182 [+0.10, +0.26] on BFCL. The reader was scored on value errors on four runs, the three Llama-3.2-1B runs and Llama-3.2-3B on Glaive, and is within noise of the probe on all of them, with probe minus reader from -0.051 to +0.017. On Llama-3.2-3B on Glaive the output-only judge recovers none of the probe's lead and the reader matches the probe (0.951 against 0.926), so on that run the value errors are visible to a second model and not to n-gram features.

On every scored failure. On the probe side the output-only judge recovers 57 to 95% of the probe's lead on four of

Table 1. Every run under the corrected labels. n_+ and n_- are the labelled failures and valid calls in the scored population. The AUC columns are the length floor, the mean log-probability (conf.), the output-only judge, the Qwen3.5-0.8B reader and the token-role probe, all on every scored failure. The last four columns are differences of AUC on held-out tools: probe minus confidence on every scored failure with its 95% tool-resampled interval, probe minus the output-only judge on every scored failure, and probe minus confidence on wrong argument values and on dropped parallel calls within the parallel categories, each against valid calls. $^\circ$ the interval excludes zero (intervals in Sections F and G). n.c. not computed: too few failures of that kind or of valid parallel calls, an underpowered run, or a comparator not fitted on that run. * re-extracted under the corrected extractor. † fewer than thirty items of one class.

Model	Data	n_+	n_-	floor	conf.	judge	reader	probe	probe – conf.	probe – judge	values	dropped
Llama-3.2-1B	Glaive	221	359	0.539	0.576	0.910	0.949	0.963	+0.387 [+0.18, +0.59]	+0.053	+0.357 $^\circ$	n.c.
Llama-3.2-1B*	BFCL	161	339	0.635	0.651	0.755	0.818	0.831	+0.180 [+0.09, +0.27]	+0.077 $^\circ$	+0.240 $^\circ$	n.c.
Llama-3.2-1B	BFCL-live	208	157	0.709	0.663	0.682	0.795	0.782	+0.119 [−0.00, +0.24]	+0.101 $^\circ$	−0.010	n.c.
Llama-3.2-3B	Glaive	90	555	0.522	0.785	0.684	0.931	0.911	+0.126 [+0.02, +0.28]	+0.227	+0.145 $^\circ$	n.c.
Llama-3.2-3B	BFCL	120	576	0.618	0.707	0.633	0.758	0.831	+0.125 [+0.05, +0.20]	+0.199 $^\circ$	+0.134 $^\circ$	+0.016
Gemma-3-1B	Glaive	158	460	0.704	0.650	0.872	0.853	0.883	+0.233 [+0.11, +0.34]	+0.011	+0.226 $^\circ$	n.c.
Gemma-3-1B*	BFCL	364	150	0.842	0.711	0.900	0.927	0.948	+0.237 [+0.19, +0.29]	+0.048 $^\circ$	+0.158	n.c.
Qwen3-1.7B	BFCL	52	639	0.628	0.799	0.558	0.662	0.745	−0.053 [−0.16, +0.05]	+0.187 $^\circ$	−0.124	n.c.
Qwen3-1.7B † *	Glaive	10	699	0.677	0.867	n.c.	n.c.	0.676	−0.191 [−0.65, +0.21]	n.c.	n.c.	n.c.
MiniCPM5-2B*	BFCL	54	640	0.578	0.823	0.585	0.677	0.775	−0.048 [−0.15, +0.05]	+0.190 $^\circ$	−0.162 $^\circ$	n.c.
MiniCPM5-2B*	BFCL-live	120	475	0.545	0.797	0.642	0.655	0.744	−0.053 [−0.15, +0.02]	+0.102 $^\circ$	−0.048	n.c.
MiniCPM5-2B † *	Glaive	9	716	0.685	0.913	n.c.	n.c.	0.710	−0.203 [−0.51, +0.01]	n.c.	n.c.	n.c.
Qwen3.5-0.8B*	BFCL	86	398	0.601	0.792	0.678	0.730	0.682	−0.110 [−0.19, −0.03]	+0.004	−0.121 $^\circ$	n.c.
MiniCPM5-2B*	BFCL, JSON	83	514	n.c.	0.757	n.c.	n.c.	0.882	+0.125 [+0.05, +0.19]	n.c.	−0.066	+0.330 $^\circ$
Qwen3.5-0.8B*	BFCL, JSON	226	436	n.c.	0.808	n.c.	n.c.	0.868	+0.060 [+0.01, +0.11]	n.c.	−0.136 $^\circ$	+0.198 $^\circ$

the seven runs, and the reader is within noise of the probe on six of seven (Figure 2). On the confidence side both score below confidence, the judge at 0.558 to 0.678 because a few dozen failures cannot fit its n-gram features.

What the probe adds. On most probe-side runs the probe’s lead over confidence is visible in the call text, so what it shows is that the error can be read from the output. The reading that the model’s states hold what its text does not would need a run where the probe beats both output-side comparators on value errors, and on none of the four runs where both were scored does it. Fusing the probe with the output-only judge by an equal-weight rank average gains +0.021 to +0.161 over the judge alone on every scored failure on the probe side, with an interval excluding zero on five of seven runs.

6. Controls on the split

A difference between model families can come from the items, the labels, the prompt or the extraction. Each paragraph below changes one thing and keeps the rest of the protocol fixed. The difficulty, budget and summary controls were run on every scored failure, before the decomposition by kind of error.

Item difficulty. A model that fails rarely fails on hard items, and its confidence may register hardness instead of its own error. On BFCL an item’s difficulty is the share of the other checkpoints that fail it. That share alone predicts a model’s failures at 0.697 to 0.782 AUC. Scoring both de-

tectors inside strata of difficulty removes what either gains from knowing which items are hard, and the split does not close (Figure 3a). The advantage is +0.182 on Llama-3.2-1B and +0.182 on Llama-3.2-3B, and between −0.099 and −0.062 on the three confidence-side checkpoints. Confidence on the confidence side tracks the model’s own errors beyond the hardness of the request.

Label budget. A probe needs labelled failures and confidence needs none, and the confidence-side models fail less often. Training the Llama-3.2 probes on as few failures as the other families supply, with the test folds untouched, costs them 0.041 AUC at 1B and 0.018 at 3B (Figure 3b). At 51 training failures the advantage is still +0.137 and +0.110, above every value on the other side.

The confidence summary. The mean log-probability averages over every token of the call, while the probe reads the argument values separately, so a summary restricted to the value tokens could do better. Only two of the seven probe-side runs store several summaries, three of them over the call’s own tokens, and none stores one over the value tokens alone. On these two runs the summary with the highest AUC on the other folds, applied fold by fold, moves the advantage to +0.165 on Llama-3.2-1B and +0.330 on Gemma-3, and an oracle that picks the best summary on the test items leaves +0.165 and +0.237 (Figure 3c, Section I). On the confidence side the fold-chosen advantage is at most −0.032.

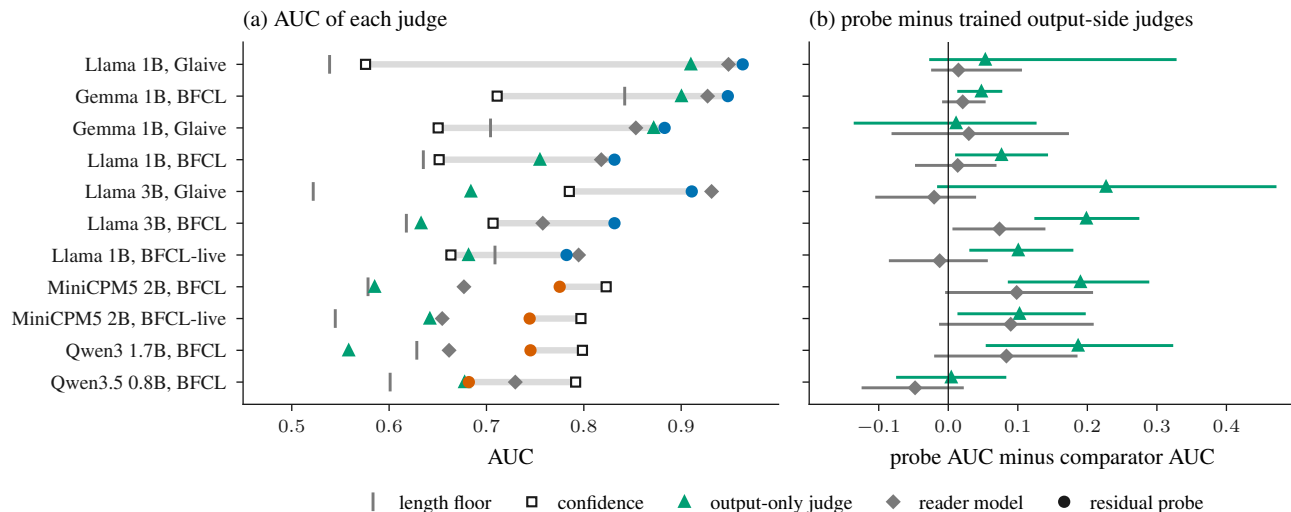


Figure 2. Every scored failure. (a) AUC per powered run of the length floor, the mean log-probability, the output-only judge, the Qwen3.5-0.8B reader and the token-role probe, with the span from confidence to probe shaded. (b) Probe AUC minus output-only judge AUC (triangles) and minus reader AUC (diamonds), with 95% tool-resampled intervals.

Table 2. Wrong argument values against valid calls, per run. Probe minus confidence (point estimate), probe minus the output-only judge, the share of the probe’s lead over confidence the judge recovers (n.c. where the probe does not lead), probe minus reader and reader minus confidence, with 95% tool-resampled intervals.

Run	probe – conf.	probe – judge	judge share	probe – reader	reader – conf.
Llama-3.2-1B, Glaive	+0.357	+0.085 [−0.04, +0.51]	76%	+0.017 [−0.05, +0.18]	+0.339 [−0.14, +0.69]
Llama-3.2-1B, BFCL	+0.240	+0.086 [+0.01, +0.16]	64%	+0.015 [−0.06, +0.08]	+0.225 [+0.11, +0.33]
Llama-3.2-1B, BFCL-live	−0.010	+0.162 [+0.05, +0.27]	n.c.	−0.051 [−0.17, +0.07]	+0.040 [−0.07, +0.16]
Llama-3.2-3B, Glaive	+0.145	+0.225 [−0.03, +0.48]	0%	−0.026 [−0.11, +0.02]	+0.170 [+0.08, +0.32]
Llama-3.2-3B, BFCL	+0.134	+0.182 [+0.10, +0.26]	0%	n.c.	n.c.
Gemma-3-1B, Glaive	+0.226	+0.013 [−0.23, +0.24]	94%	n.c.	n.c.
Gemma-3-1B, BFCL	+0.158	+0.098 [−0.07, +0.22]	38%	n.c.	n.c.
Qwen3-1.7B, BFCL	−0.124	+0.119 [−0.04, +0.28]	n.c.	n.c.	n.c.
MiniCPM5-2B, BFCL	−0.162	+0.151 [−0.04, +0.32]	n.c.	n.c.	n.c.
MiniCPM5-2B, BFCL-live	−0.048	+0.089 [−0.03, +0.21]	n.c.	n.c.	n.c.
Qwen3.5-0.8B, BFCL	−0.121	+0.020 [−0.07, +0.11]	n.c.	n.c.	n.c.

Benchmark and schema echoes. On the probe side, Gemma-3 reads +0.233 on Glaive and +0.237 on BFCL and Llama-3.2-1B keeps its sign on three benchmarks. The confidence side has powered runs on BFCL and BFCL-live only, so its benchmark breadth is two. Keeping the first occurrence of each unique Glaive request gives +0.389 on Llama-3.2-1B, +0.246 on Gemma-3 and +0.190 on Llama-3.2-3B. With matched schema echoes removed from evaluation, the pooled advantage keeps its sign on six of the seven probe-side runs. The exception is Llama-3.2-1B on the live categories, which moves from +0.119 to −0.020 [−0.16, +0.10]. Its value-error advantage is also near zero, so this run is on neither side once format failures are set aside.

Extraction provenance. The first extractor produced five of the seven probe-side runs and one of the four confidence-side runs, so extractor version lines up with side. The one run extracted under both reads +0.224 and +0.180, a drift

of 0.044, small against the probe-side advantages and not small against single-run differences on the confidence side. Differences below this drift are not interpreted.

The permutation null of the probe pipeline, the fold-level gaps and the protocol choices that move the numbers are in Section J.

7. Attention readouts, head averaging and cost

A stack may expose attention weights but not activations. Three readouts of the attention on the generated call are compared here. The per-head spectral profile takes each head’s attention on the call, symmetrises it into a graph and keeps five eigenvalue statistics of its normalised Laplacian per head and layer. The head-averaged profile computes the same statistics on the graph averaged over heads, which is how most of this literature reads a layer. The anchored

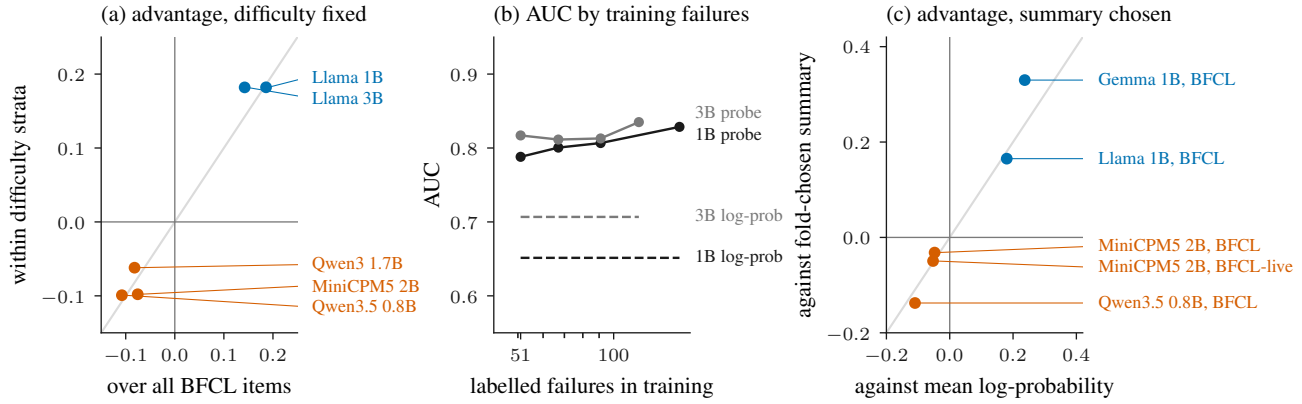


Figure 3. (a) Internal advantage within strata of item difficulty against the advantage over all BFCL items, one point per checkpoint, identity line grey. Difficulty is the share of the other checkpoints that fail the item. (b) Token-role probe AUC (solid) and mean log-probability AUC (dashed) against the labelled failures available for training on Llama-3.2-1B and 3B, log scale. (c) Internal advantage against a confidence summary chosen on training folds and against the mean log-probability, one point per run that stores several summaries, identity line grey.

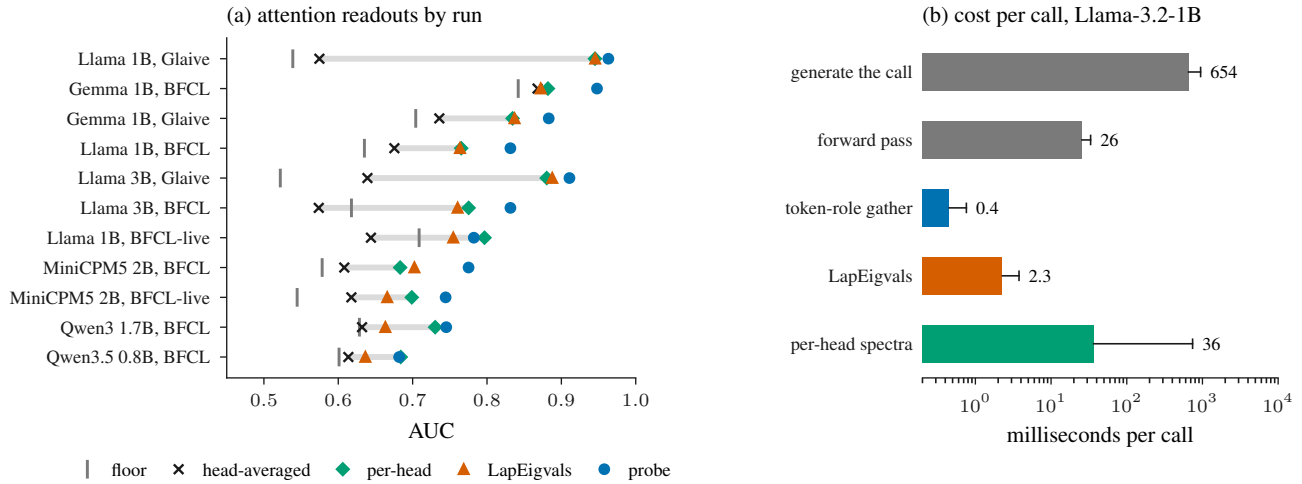


Figure 4. (a) AUC per powered run, every scored failure, of the length floor, the head-averaged spectral profile, the per-head spectral profile, LapEigvals and the token-role probe, with the span from head-averaged to per-head shaded. (b) Cost per call of each path on Llama-3.2-1B, bar at the median over 20 BFCL prompts, line to the 90th percentile, log scale, one consumer GPU.

readout keeps the attention rows of the call’s own tokens per head, without a spectrum.

On the probe-side runs the per-head profile and LapEigvals trail the token-role probe by a few points (Figure 4a), for instance $+0.066$ on Llama-3.2-1B and $+0.066$ on Gemma-3 on BFCL. The two are within noise of each other, and on Llama-3.2-1B and Gemma-3 on BFCL, the two runs where it was computed, the anchored readout ties the per-head profile ($+0.007$ and $+0.037$).

Head averaging. The head-averaged profile sits within -0.065 to $+0.117$ of the length floor on the probe-side runs. Keeping heads apart gains $+0.081$ on average over 11 runs, and permuting the head axis per item, which keeps every value and destroys only which head produced it, costs

$+0.195$ (Sections A and K).

Cost. Reading confidence costs nothing beyond generation. Every internal judge needs a teacher-forced pass over prompt and call, 26 ms against 654 ms of generation (Figure 4b). On top of the pass the token-role features cost 0.4 ms and LapEigvals 2 ms, and the per-head spectra 36 ms at the median and 741 ms at the 90th percentile. The output-only judge needs no pass of the deployed model, and the reader one pass of Qwen3.5-0.8B over the request and the call.

8. Discussion

Readings of the split. Three properties of the model set fit all six checkpoints (Table 4), and the data do not separate them. The first is release date, with a cutoff between March and April 2025. The second is a reasoning mode in post-training: the three confidence-side families ship a chat template with a reasoning mode, and the two probe-side families do not. Post-training that rewards a model for working through a call before writing it may leave its token probabilities tracking its own value errors. This reading was written down as hypothesis H1 after all six checkpoints had been scored (Section D), so it is a prediction for new models and not a test on these. The third is native tool-calling post-training: Gemma-3 ships no tool template and is prompted through the fallback, and Yeats et al. (2026) find that tool-specific fine-tuning changes what a probe reads, so how much tool-call data a model saw in post-training could move the side as well. Switching the reasoning mode on at generation within Qwen3-1.7B yields 21 failures in 396 scored calls, too few to score, and Qwen3 is on the confidence side with the mode off.

What confidence misses. On the probe side the mean log-probability ranks wrong values close to chance on BFCL, 0.587 on Llama-3.2-1B and 0.552 on Gemma-3, while the residual stream separates them and, on most runs, so does a reader of the call. We interpret this as an output distribution that does not express what the model’s own text shows. A confidence summary over the value tokens alone was not stored, so part of the miss may belong to averaging over the whole call (Section 6).

For operators. The decision that changes is which judge to build, if any, in order of cost.

Before building an internal judge for a tool-calling model

1. Score the mean token log-probability on wrong argument values, on a few hundred labelled calls from the model to be deployed, on tools absent from the labelled set. Where it already ranks them (Table 1, confidence side), a probe fused with it gained at most +0.034 AUC on every scored failure.
2. Where it misses them, train an output-only judge or a small reader model on the same labels before reading internals. On value errors the judge recovered at least half of the probe’s lead on three of six such runs, and none on Llama-3.2-3B, so check the trained judge against a probe on the deployed model before choosing.
3. For parallel requests, compare the number of calls written with the number the request asks for. A one-bit category flag already separates dropped calls from valid ones on every run with enough of them, and a probe adds nothing measurable within the parallel categories on the one run prompted through its native template.

For detector research. A probe’s accuracy reported without the model’s own confidence, a trained output-only judge and held-out tools can show a strong detector on a model that did not need one: here confidence reaches 0.576 to 0.823 AUC and beats the probe on value errors on every Qwen3, Qwen3.5 and MiniCPM5 run.

9. Conclusion

On wrong argument values, a probe on the residual stream beats the mean token log-probability on Llama-3.2 and Gemma-3 and trails it on Qwen3, Qwen3.5 and MiniCPM5, including when the latter are made to write JSON. This is a description of three checkpoints per side, assigned by family, at three billion parameters or fewer. On most probe-side runs the probe’s lead is also visible to a judge that reads only the output. Two recommendations follow, in order of cost: score the model’s confidence on its value errors on held-out tools before building any judge, and where it misses them, train a judge on the output text and compare it with a probe before choosing. A third is cheaper than either: for parallel requests, count the calls.

10. Limitations

The first limitation is the instrument. Labels come from comparing a call with a benchmark’s ground truth, which is a proxy for whether the action would have been harmful, and Section C shows that a parser can manufacture a family effect. The hand audit after the second repair found 18 errors in 240 items, all correct calls scored as failures, and the residual after the third repair is unmeasured. Errors of that kind favour probes, so residual label error would move runs toward the probe side.

The second is the design. The sides were assigned by family after the split had been seen, three checkpoints each, so no checkpoint-level test can separate them, and release date, a reasoning mode and native tool-calling post-training all fit the division. The prompt differs by family: Gemma-3 and the forced-JSON runs go through a fallback prompt that asks for one JSON object. The comparator favours the probe, which is trained, over confidence, which is not, and the reader sees no tool schemas, so it understates what the text reveals.

The third is provenance. The first extractor produced five of the seven probe-side runs and one of the four confidence-side runs and recorded no commit, and six of the re-extracted runs, among them three confidence-side runs, come from a tree with uncommitted changes (Table 5). A run extracted twice drifted by 0.044.

The fourth is scale and realism. Every checkpoint has at most three billion parameters, and Yeats et al. (2026) find

probe accuracy rising with size, so the side a family falls on above this range is open. Calls are single-step on three benchmarks of English requests, the confidence side is powered on two of them, the multi-turn replay is one anecdotal run, and cost is measured on one consumer GPU.

Impact statement

This work measures open-weight language models on public tool-calling benchmarks, trains no language model and involves no human subjects. Its aim is defensive: to tell an operator which wrong tool calls a model’s own confidence already flags and which ones need another judge before the call executes. The same detectors could guide the construction of calls that evade them, which is why we report every kind of error and every run on which a detector weakens rather than one headline number.

Reproducibility statement

Every number in this paper expands from a macro that a released script builds from the result files, and a provenance file records the file, field and run each macro reads. Labels are recomputed from the stored generations at evaluation time, and relabelling the stored generations of 16 runs with the released labeller reproduces every stored label. Splits are grouped by ground-truth tool and every interval resamples tools. The extractions themselves are not uniform: Table 5 lists, per run, the extractor generation, the commit and whether the tree had uncommitted changes, and the first-generation runs record none of these. Section D dates each registered prediction against the data that test it. Models and benchmarks are public, and the stored extractions the tables are computed from are released with the code.

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A. Head averaging under causal attention

Purpose: two elementary facts behind the head-averaging measurement of Section 7, stated for causal attention. Neither predicts which models fall on which side of the value-error split, and neither is a contribution of this paper.

Fix one layer with H heads and T positions. Head h produces a row-stochastic attention matrix A_h , lower triangular under a causal mask, symmetrised and stripped of self-loops into $W_h = \frac{1}{2}(A_h + A_h^\top) - \text{diag}(\frac{1}{2}(A_h + A_h^\top))$, with combinatorial Laplacian $L_h = D_h - W_h$ and symmetric normalised Laplacian $\mathcal{L}_h = I - D_h^{-1/2}W_hD_h^{-1/2}$. The head-averaged graph is $\bar{W} = \frac{1}{H} \sum_h W_h$, and λ_2 is the second-smallest eigenvalue.

Remark 1 (head averaging is many-to-one). Any map of the form $\{W_h\} \mapsto \Phi(\bar{W})$ is constant on head configurations sharing the same mean, because it factors through $\bar{W}$. For $T \geq 4$ two configurations of two causal row-stochastic heads with the same mean exist whose spreads of per-head connectivity, $\max_h \lambda_2(L_h) - \min_h \lambda_2(L_h)$, differ by $\cos(\pi/T) - \frac{1}{2}$, which is at least $\cos(\pi/4) - \frac{1}{2} \approx 0.21$. Let A_P be the previous-token head, $A_P[i, i-1] = 1$ for $i \geq 1$ and $A_P[0, 0] = 1$, and A_S the sink head, $A_S[i, 0] = 1$ for every i . Both are causal and row-stochastic, and so is $A_M = \frac{1}{2}(A_P + A_S)$. Symmetrisation and the removal of the diagonal are linear, so $\{A_P, A_S\}$ and $\{A_M, A_M\}$ share the mean graph. W_P is the path on T nodes with edge weights $\frac{1}{2}$, so $\lambda_2(L_P) = 1 - \cos(\pi/T)$, and W_S the star on position 0 with edge weights $\frac{1}{2}$, so $\lambda_2(L_S) = \frac{1}{2}$ for $T \geq 3$. The spread is $\cos(\pi/T) - \frac{1}{2}$ in the first configuration and zero in the second.

Remark 2 (combinatorial connectivity of the averaged graph). For any heads, $\lambda_2(\bar{L}) \geq \frac{1}{H} \sum_h \lambda_2(L_h)$, with equality if and only if the heads admit a common Fiedler vector. The Laplacian is linear in the adjacency, and by the min-max theorem of Courant and Fischer λ_2 is a minimum of linear functionals of L and hence concave (Chung, 1997).

Scope. The second remark covers the combinatorial Laplacian only. The spectral readouts of Section 7 use the normalised Laplacian, whose Rayleigh quotient is not linear in W , and for it no inequality in either direction holds. In the construction of the first remark, computed by `analysis/v2-causal-construction.py`, the normalised connectivity of the averaged graph minus the per-head mean is $+0.090$ at $T = 5$ and -0.166 at $T = 200$, and the combinatorial difference is $+0.116$ and $+0.0001$. On 210 measured layers from three models and three inputs the combinatorial inequality holds without violation, the normalised one fails on three layers, and the averaged graph reports connectivity a median 14% above the mean over heads.

Remark 3 (LapEigvals reads degrees). Binkowski et al. (2025) observe that for a causal attention matrix the Laplacian $L = D - A$ is lower triangular, so its spectrum is its diagonal $\{d_j - A_{jj}\}$, and their top- k features depend on A only through its column sums and diagonal. The symmetrisation above destroys triangularity, so the two readouts are not two estimates of one object, and what they share is that both keep heads apart.

B. Protocol details

Purpose: every setting needed to rerun the extraction and the scoring without another paper.

Generation. Greedy decoding in bfloat16 with deterministic kernels, a budget of 256 new tokens (160 on multi-turn prompts), tool schemas passed through the model’s chat template with its reasoning mode disabled, and every end-of-turn id of the template among the stopping ids. The rendered text is tokenised without a second beginning-of-sequence token on the

re-extracted runs. Runs from the corrected extractor store their budget, library versions, commit and a flag for uncommitted changes (Table 5). Runs from the first extractor store none of these.

Fallback prompt. When a model’s chat template drops the tool argument (Gemma-3), or under forced JSON, the schemas are passed in this system prompt, followed by one JSON schema per line: “You are a helpful assistant with access to the following functions. When a function is needed, reply with ONLY a JSON object of the form `{“name”: <function name>, “arguments”: {<arg name>: <value>}}` and nothing else. Available functions:”. It asks for one object, which a model can read as a request to write one call (`run_pilot_v2.py`, `TOOL_PROMPT_FALLBACK`).

Features. Residual states are read at eight depths spaced evenly over the layer count at the function-name token, the mean over argument-value tokens and the closing delimiter, located by aligning the parsed call to token offsets in each dialect. Attention features are reduced inside a forward hook on each attention module, on the subgraph induced on the generated call.

Output-only judge and reader. The structural features are the floor’s three lengths, a one-hot benchmark category, and counts read off the generated call: calls, arguments, characters, digits, quoted strings and schema keywords. The confidence features are every stored summary of the token log-probabilities. The text features are character 2 to 5-gram TF-IDF vectors of the call and of the request. The output-only judge concatenates all three. The reader model Qwen3.5-0.8B reads the request, a newline and the generated call, without the tool schemas, and its residual state at eight depths is read at the last token and averaged over the call’s tokens. Both are fitted with the probe’s classifier on the probe’s folds, training population and validation carve-out, and the folds are asserted equal to the stored ones. As a reverse check the same code recomputes the paper’s length floor, which matches the stored floor on every run to within $4e-3$.

Classifiers and scoring. Logistic regression with ℓ_2 penalty, standardised features, class balancing and $C \in \{0.01, 0.1, 1, 10\}$ chosen on a one-fifth validation carve-out of training tools. Five tool-grouped folds, training restricted to the scored population, one pooled AUC per seed over five seeds, mean reported. Differences are paired bootstraps that resample whole tools, 2000 draws per seed for the stored contrasts and 1000 for the audit contrasts, draws pooled over seeds, percentile intervals.

C. Labelling and extraction defects found by the audit

Purpose: the repairs that condition every number, with the side each defect favoured.

Defect	Where it fell	Favoured
A predicted list argument was collapsed to its first element by the unwrapping meant for ground truth, so correct list-valued calls were scored wrong	JSON formats (Llama-3.2, Qwen3). 158 → 52 failures on Qwen3-1.7B	probes, through a surface feature
Semicolon-separated parallel calls did not parse and were scored unparseable	Llama-3.2-3B, 185 calls returned to the scored population	probes
A list sent as a string was compared inconsistently with a list	Llama-3.2	probes
Probe positions never located in the MiniCPM5 dialect, so the probe read the end-of-sequence state for every role	MiniCPM5	confidence
Generation stopped only on end-of-text, so Qwen3.5 ran past its end of turn	Qwen3.5	unclear, small
A second beginning-of-sequence token	Llama-3.2, MiniCPM5	neither
Probes trained on every item and scored on a subset learned format failures the subset excludes	population including abstentions	probes
The sign of the log-probability fixed on items it is never scored on	Llama-3.2, two runs	probes
Runs sharing a checkpoint treated as independent in a rank test	aggregate	the apparent significance

Table 3 reports a hand-checked sample of thirty items per run drawn on the labels after the first two repairs, judged against the benchmark’s function documents. Every error found was a correct call scored as a failure, 18 of 240 sampled items, from two causes the benchmark’s own checker handles: list arguments written as single-quoted Python literals, and mathematical

expressions that differ only in notation. Both are repaired in the released labeller, which applies the checker’s string normalisation to both sides, and every number in this paper is computed under that third repair. The residual error after the third repair has not been sampled by hand. Errors of the kind found inflate probes, which can learn the surface form of a mislabelled correct call.

Table 3. Hand audit of the labels after the first two repairs, thirty items per run stratified over failure modes, Wilson intervals. All errors are correct calls scored as failures.

Model	Data	n	errors	rate [95% interval]	dominant cause
Llama-3.2-1B	BFCL	30	5	0.17 [0.07, 0.34]	python literal not decoded
Llama-3.2-3B	BFCL	30	6	0.20 [0.10, 0.37]	python literal not decoded
Qwen3-1.7B	BFCL	30	2	0.07 [0.02, 0.21]	notation caret
Gemma-3-1B	BFCL	30	0	0.00 [0.00, 0.11]	none
MiniCPM5-2B	BFCL	30	4	0.13 [0.05, 0.30]	notation caret
Qwen3.5-0.8B	BFCL	30	1	0.03 [0.01, 0.17]	notation caret
Llama-3.2-1B	Glaive	30	0	0.00 [0.00, 0.11]	none
Gemma-3-1B	Glaive	30	0	0.00 [0.00, 0.11]	none

D. Registered predictions and outcomes

Purpose: every prediction committed to the repository, beside its outcome. C1 was committed before the audit, R1 to R3 before the first labelling repair, and H1 and H2 after the first two repairs and before the third. R1 and H1 were written when all six checkpoints had already been scored, R1 under the unrepaired labels, so neither is a test on these checkpoints. P1 and P2 were written on the day of the October audit, before the measurement.

Id	Prediction	Outcome
R1	The internal advantage separates the families by release period: MiniCPM5 and Qwen3.5 below Llama-3.2, Gemma-3 and Qwen3 at every checkpoint.	Failed. Qwen3-1.7B moved to the confidence side once the list-argument defect was repaired. The division that holds was found afterwards and is descriptive.
R1b	The tool-tuned Llama-xLAM-2-8b has a smaller internal advantage than Llama-3.1-8B-Instruct, with an interval excluding zero.	Not run.
R2	On a distractor-tool dose ladder the internal advantage rises with the number of distractors on at least one model from each side.	Not run.
R3	Under forced JSON output the families where confidence wins keep an advantage at or below zero, and the prediction fails if it exceeds +0.10.	Failed on MiniCPM5-2B, +0.125 [+0.05, +0.19], on a partial run of 616 of 850 items extracted from a tree with uncommitted changes. Undecided on Qwen3.5-0.8B, +0.060 [+0.01, +0.11]. Dropped parallel calls, which the fallback prompt invites, make up most failures on both forced runs. On wrong argument values both stay below zero (Section 4).
C1	The per-head spectral score and LapEigvals are not interchangeable, each keeps signal when the other is conditioned on, and their union exceeds both.	Two parts held and the third failed (union gain within noise on eleven runs), measured on the labels before the audit.
H1	Models whose post-training included a reasoning mode have an internal advantage lower by at least 0.05 than a matched model without it, same base.	No matched pair has been run. Qwen3-1.7B with its reasoning mode switched on at generation fails on 21 of 396 scored calls, below the thirty needed to score.
H2	Within one model the advantage stays within 0.05 of its full value when its labelled failures are cut to what the other families supply and when distractor tools are added.	The label half holds (Figure 3b). The distractor half is not run.
P1	On dropped parallel calls, the entropy of the step at which the model stops beats the mean log-probability on every run with enough of them.	Failed. It held on Gemma-3, +0.155, and on Qwen3.5-0.8B under forced JSON, +0.057, and went the other way on MiniCPM5-2B under forced JSON, −0.214.
P2	On wrong argument values the stop-step entropy adds nothing over the mean log-probability.	Held on seven of seven runs.

E. Every detector on every run

Purpose: the AUC of every detector, so that any comparison in the body can be recomputed. SinkProbe and the anchored readout were computed only on runs extracted after these detectors were added, and the other cells read n.c.

Model	Data	token-role	token-level	head-avg.	per-head	LapEigvals	Lookback	SinkProbe	anchored	log-prob	floor
Llama-3.2-1B	Glaive	0.963	0.972	0.575	0.945	0.945	0.943	n.c.	n.c.	0.576	0.539
Llama-3.2-1B	BFCL	0.831	0.798	0.675	0.765	0.764	0.758	0.789	0.773	0.651	0.635
Llama-3.2-1B	BFCL-live	0.782	0.765	0.644	0.797	0.755	0.808	n.c.	n.c.	0.663	0.709
Llama-3.2-3B	Glaive	0.911	0.861	0.639	0.880	0.888	0.892	n.c.	n.c.	0.785	0.522
Llama-3.2-3B	BFCL	0.831	0.786	0.574	0.775	0.760	0.753	n.c.	n.c.	0.707	0.618
Gemma-3-1B	Glaive	0.883	0.898	0.736	0.834	0.837	0.786	n.c.	n.c.	0.650	0.704
Gemma-3-1B	BFCL	0.948	0.934	0.868	0.882	0.872	0.923	0.919	0.919	0.711	0.842
Qwen3-1.7B	BFCL	0.745	0.655	0.632	0.730	0.663	0.711	n.c.	n.c.	0.799	0.628
Qwen3-1.7B [†]	Glaive	0.676	0.790	0.445	0.781	0.678	0.709	0.702	0.674	0.867	0.677
MiniCPM5-2B	BFCL	0.775	0.775	0.608	0.683	0.702	0.744	0.754	0.764	0.823	0.578
MiniCPM5-2B	BFCL-live	0.744	0.742	0.618	0.699	0.666	0.732	0.706	0.700	0.797	0.545
MiniCPM5-2B [†]	Glaive	0.710	0.760	0.635	0.713	0.652	0.723	0.690	0.773	0.913	0.685
Qwen3.5-0.8B	BFCL	0.682	0.717	0.614	0.684	0.636	0.680	0.708	0.756	0.792	0.601

F. Output-only judges and the reader model per run

Purpose: each output-side comparator of Section 5 on each run, with its paired interval against the probe. Structural floor, confidence summaries in one regression and text n-grams are the three parts of the output-only judge. The fusion is an equal-weight rank average of the probe and the output-only judge, with no fitted weight. All on every scored failure. The reader is Qwen3.5-0.8B, the same checkpoint as one confidence-side model.

Model	Data	struct.	conf. LR	text	judge	probe — judge	fusion — judge	reader	probe — reader
Llama-3.2-1B	Glaive	0.557	0.493	0.916	0.910	+0.053 [-0.03, +0.33]	+0.042 [-0.00, +0.23]	0.949	+0.015 [-0.02, +0.11]
Llama-3.2-1B	BFCL	0.684	0.682	0.704	0.755	+0.077 [+0.01, +0.14]	+0.066 [+0.03, +0.10]	0.818	+0.014 [-0.05, +0.07]
Llama-3.2-1B	BFCL-live	0.685	0.653	0.680	0.682	+0.101 [+0.03, +0.18]	+0.075 [+0.03, +0.12]	0.795	-0.012 [-0.09, +0.06]
Llama-3.2-3B	Glaive	0.415	0.785	0.671	0.684	+0.227 [-0.02, +0.47]	+0.161 [+0.03, +0.32]	0.931	-0.020 [-0.11, +0.04]
Llama-3.2-3B	BFCL	0.681	0.702	0.627	0.633	+0.199 [+0.12, +0.27]	+0.142 [+0.10, +0.18]	0.758	+0.074 [+0.01, +0.14]
Gemma-3-1B	Glaive	0.599	0.519	0.869	0.872	+0.011 [-0.14, +0.13]	+0.021 [-0.05, +0.08]	0.853	+0.030 [-0.08, +0.17]
Gemma-3-1B	BFCL	0.916	0.856	0.882	0.900	+0.048 [+0.01, +0.08]	+0.039 [+0.02, +0.06]	0.927	+0.021 [-0.01, +0.05]
Qwen3-1.7B	BFCL	0.662	0.735	0.551	0.558	+0.187 [+0.05, +0.32]	+0.131 [+0.07, +0.19]	0.662	+0.084 [-0.02, +0.19]
MiniCPM5-2B	BFCL	0.641	0.779	0.557	0.585	+0.190 [+0.09, +0.29]	+0.131 [+0.08, +0.19]	0.677	+0.098 [-0.00, +0.21]
MiniCPM5-2B	BFCL-live	0.507	0.756	0.623	0.642	+0.102 [+0.01, +0.20]	+0.091 [+0.03, +0.15]	0.655	+0.090 [-0.01, +0.21]
Qwen3.5-0.8B	BFCL	0.663	0.838	0.660	0.678	+0.004 [-0.08, +0.08]	+0.031 [-0.01, +0.08]	0.730	-0.048 [-0.12, +0.02]

G. Every kind of error on every run

Purpose: the internal advantage of each kind of error against valid calls, on every run with at least twenty failures of that kind, and dropped calls scored within the parallel categories where both classes have twenty items. Wrong tool names reach twenty only on Llama-3.2-1B on BFCL-live, 68 of them, with an advantage of +0.216 [+0.06, +0.36].

Run	n_+	wrong values	n_+	missing arguments	n_+	dropped calls	dropped, parallel only
Llama-3.2-1B, Glaive	132	+0.357 [+0.00, +0.68]	86	+0.441 [+0.27, +0.63]	n.c.	n.c.	n.c.
Llama-3.2-1B, BFCL	127	+0.240 [+0.14, +0.34]	21	-0.055 [-0.18, +0.04]	n.c.	n.c.	n.c.
Llama-3.2-1B, BFCL-live	89	-0.010 [-0.20, +0.16]	46	+0.223 [+0.09, +0.36]	n.c.	n.c.	n.c.
Llama-3.2-3B, Glaive	85	+0.145 [+0.05, +0.32]	n.c.	n.c.	n.c.	n.c.	n.c.
Llama-3.2-3B, BFCL	88	+0.134 [+0.05, +0.22]	n.c.	n.c.	25	+0.158 [+0.05, +0.27]	+0.016 [-0.10, +0.13]
Gemma-3-1B, Glaive	109	+0.226 [+0.03, +0.36]	47	+0.260 [+0.09, +0.48]	n.c.	n.c.	n.c.
Gemma-3-1B, BFCL	33	+0.158 [-0.04, +0.33]	219	+0.188 [+0.14, +0.24]	99	+0.342 [+0.27, +0.42]	n.c.
Qwen3-1.7B, BFCL	33	-0.124 [-0.25, +0.00]	n.c.	n.c.	n.c.	n.c.	n.c.
MiniCPM5-2B, BFCL	22	-0.162 [-0.30, -0.02]	n.c.	n.c.	n.c.	n.c.	n.c.
MiniCPM5-2B, BFCL-live	84	-0.048 [-0.16, +0.04]	24	-0.085 [-0.28, +0.05]	n.c.	n.c.	n.c.
Qwen3.5-0.8B, BFCL	54	-0.121 [-0.22, -0.03]	n.c.	n.c.	n.c.	n.c.	n.c.
MiniCPM5-2B, BFCL, JSON	21	-0.066 [-0.31, +0.12]	n.c.	n.c.	56	+0.201 [+0.15, +0.25]	+0.330 [+0.20, +0.45]
Qwen3.5-0.8B, BFCL, JSON	64	-0.136 [-0.23, -0.04]	n.c.	n.c.	143	+0.171 [+0.13, +0.21]	+0.198 [+0.09, +0.32]

H. Worked examples of each kind of error

Purpose: one stored generation per kind of error, so a reader can see what each label means. Each row is the first call-required item in file order with that failure mode, with two filters: the second row is the first whose call matches the schema-echo pattern of Section 3, and every other row skips calls that match the pattern or contain a `description` field, which copies

part of a schema. A partial echo of that second kind is not matched by the pattern and can be labelled a value error, so the echo shares of Section 3 are lower bounds.

Model	Mode	Ground truth	Generated call
Gemma-3-1B	wrong arg values	[{"calculate_resonant_frequency": {"inductance": [0.05], "capacitance": [0.0001], "round_off": ["", 2]}}]	```json {"name": "calculate_resonant_frequency", "arguments": { "inductance": 50, "capacitance": 100, "round_off": 2 }}```
Gemma-3-1B	missing args	[{"algebra_quadratic_roots": {"a": [1], "b": [-3], "c": [2]}}]	{"name": "algebra.quadratic.roots", "arguments": { "type": "object", "properties": { "a": { "type": "integer", "description": "Coefficient of x^2."}, "b": { "type": "integer", "description": "Coefficient of x"}, "c": { "type": "integer", "description": "Constant term"} }}}
Gemma-3-1B	missing calls	[{"spotify_play": {"artist": ["Taylor Swift"], "duration": [20]}}, {"spotify_play": {"artist": ["Maroon 5"], "duration": ...}}]	```json {"name": "spotify.play", "arguments": { "artist": "Taylor Swift", "duration": 20, "duration": 15 }}```
Qwen3.5-0.8B, forced JSON	missing calls	[{"spotify_play": {"artist": ["Taylor Swift"], "duration": [20]}}, {"spotify_play": {"artist": ["Maroon 5"], "duration": ...}}]	{"name": "spotify.play", "arguments": {"artist": "Taylor Swift", "duration": 120}}
Qwen3.5-0.8B	wrong arg values	[{"solve_quadratic": {"a": [3], "b": [-11], "c": [-4], "root_type": ["all"]}}]	<tool.call> <function=solve_quadratic> <parameter=a> 3 </parameter> <parameter=b> -11 </parameter> <parameter=c> -4 </parameter> <parameter=root_type> real </pa ...<tool.call> <function=math.factorial> <parameter=number> 5 </parameter> </function> </tool.call>
Qwen3.5-0.8B	valid	[{"math_factorial": {"number": [5]}}]	

I. Confidence summaries

Purpose: the fold-chosen confidence summary of Section 6 in full.

On the re-extracted runs eleven length-invariant summaries of the token log-probabilities are stored per call: the mean, the minimum, the mean of the five least likely tokens, the perplexity, the fraction of tokens below one half, the mean, maximum and final predictive entropy, and the mean, minimum and mean-of-five over the call's own tokens with special tokens removed. For each fold the summary with the highest AUC on the items of the other folds is applied to the fold's items. On Llama-3.2-1B on BFCL the chosen summary reaches 0.666 against 0.651 for the mean, and the advantage becomes +0.165 [+0.09, +0.25]. On Gemma-3-1B on BFCL the chosen summary pools to 0.618, below the mean's 0.711, because the summary that leads on four folds trails on the fifth. On Qwen3.5-0.8B the chosen summary reaches 0.819 and the advantage -0.138 [-0.20, -0.07]. Two summaries that grow with the call's length, its token count and its total log-probability, are excluded as length features: on Gemma-3-1B on BFCL the token count alone reaches the length floor of that run, 0.842.

J. Further controls

Purpose: the chance level of the pipeline and the protocol choices that move the numbers.

Chance level of the pipeline. With the labels permuted, 20 refits of the whole probe pipeline on Qwen3.5-0.8B score 0.509 AUC with a standard deviation of 0.042. The standard deviation of the permuted advantage is 0.057, which is why single-run advantages of a few hundredths are not interpreted. Inside individual folds the advantage is negative on zero to three of 25 fold and seed pairs on the probe side and on 16 to 24 on the confidence side.

Training population and pooling. Training the probe on the scored population only, and fixing the sign of the log-probability on that population, both moved the numbers toward confidence (Section C). Pooling out-of-fold scores across folds penalises a fitted probe and not a fixed score: the advantage computed within folds exceeds the pooled one by +0.018 on average.

K. Head resolution controls

Purpose: the controls that attribute the per-head gain to head identity.

On the stored per-head features of 11 runs, under the same classifier and splits: the head-averaged graph's five metrics score

0.677, the same metrics computed per head and then averaged 0.709, all heads kept 0.790, the averaged features padded with Gaussian noise to the per-head width 0.568, and the per-head features with the head axis permuted independently per item 0.595. The gain of keeping heads apart over averaging the metrics is +0.081 and positive on 11 runs. Destroying head identity costs +0.195. Width without information changes the averaged read by -0.109 . These controls train on every item and score the call-required population, the training population of the first draft, and both arms of each contrast share it.

L. Calls with a history

Purpose: the multi-turn protocol and its anecdotal result.

The four multi-turn categories of BFCL-v4 (base, missing parameter, missing function, long context), one record per turn up to the fourth, with the ground-truth calls of the two preceding turns rendered as assistant messages, each followed by a fixed acknowledgement. In the corrupted condition one of the two preceding turns carries a call drawn from another conversation. Conversations with more than 22 tools are skipped. Folds group items that share either a tool or a conversation. Labels penalise arguments the schema does not define, because a weak model echoes tool schemas into argument-less calls on these prompts. On 538 turns of MiniCPM5-2B, 328 failures and 176 valid calls are scored. Linking tools and conversations leaves four groups, so the folds rest on four units and no interval is reported. The probe reads 0.900 and confidence 0.859. The run comes from a tree with uncommitted changes (Table 5).

M. Models and versions

Purpose: exact checkpoints and the layers read.

Table 4. The checkpoints, their release month, whether the chat template carries a reasoning mode (disabled at generation throughout) and the number of runs.

Checkpoint	Released	Reasoning mode	Runs
Llama-3.2-1B	2024-09	no	3
Llama-3.2-3B	2024-09	no	2
Gemma-3-1B	2025-03	no	2
Qwen3-1.7B	2025-04	yes	2
MiniCPM5-2B	2026-09	yes	3
Qwen3.5-0.8B	2026-02	yes	1

Checkpoint	Released	Attention	Layers read
meta-llama/Llama-3.2-1B-Instruct	2024-09	dense GQA, 16 layers, 32 heads	all
meta-llama/Llama-3.2-3B-Instruct	2024-09	dense GQA, 28 layers, 24 heads	all
google/gemma-3-1b-it	2025-03	sliding window with 4 global layers, 4 heads	all
Qwen/Qwen3-1.7B	2025-04	dense GQA, 28 layers, 16 heads	all
Qwen/Qwen3.5-0.8B	2026-02	hybrid, softmax in 6 of 24 layers	softmax layers
openbmb/MiniCPM5-2B	2026-09	dense, 42 layers, 16 heads	all

Two further Qwen3.5 checkpoints in the stored data are left out: a 2B run with too few failures to score and a 4B run extracted before the stopping-token repair. Experiments were run on a single 16 GB consumer GPU, and the reader model and all comparators on CPU. Table 5 lists, for every run, the extractor generation, the commit and whether the tree had uncommitted changes. The first extractor recorded none of these, and six of the 15 listed runs were extracted with uncommitted changes.

Table 5. Extraction provenance per run, from each run’s metadata file. “first”: the extractor before the audit, which stores no metadata. Uncommitted changes: the tree differed from the recorded commit at extraction time.

Run	Stored extraction	Extractor	Commit	Uncommitted changes
Llama-3.2-1B, Glaive	base_llama1b_glaive	first	not recorded	not recorded
Llama-3.2-1B, BFCL	v3_llama1b_bfcl	corrected	recorded	no
Llama-3.2-1B, BFCL-live	llama1b_live	first	not recorded	not recorded
Llama-3.2-3B, Glaive	base_llama3b_glaive	first	not recorded	not recorded
Llama-3.2-3B, BFCL	base_llama3b_bfcl	first	not recorded	not recorded
Gemma-3-1B, Glaive	base_gemma3_glaive	first	not recorded	not recorded
Gemma-3-1B, BFCL	v3_gemma3_1b_bfcl	corrected	recorded	no
Qwen3-1.7B, BFCL	base_qwen3_17b_bfcl	first	not recorded	not recorded
MiniCPM5-2B, BFCL	v3_minicpm5_2b_bfcl	corrected	recorded	yes
MiniCPM5-2B, BFCL-live	v3_minicpm5_2b_live	corrected	recorded	yes
Qwen3.5-0.8B, BFCL	v3_qwen35_08b_bfcl	corrected	recorded	yes
MiniCPM5-2B, BFCL, forced JSON	v3_minicpm5_2b_bfcl_json	corrected	recorded	yes
Qwen3.5-0.8B, BFCL, forced JSON	v3_qwen35_08b_bfcl_json	corrected	recorded	no
MiniCPM5-2B, BFCL multi-turn	v3_mt_minicpm	corrected	recorded	yes
Qwen3-1.7B, BFCL, reasoning mode on	v3_qwen3_17b_bfcl_think	corrected	recorded	yes